

Online Calibration of MATSim Models via Stochastic Batch Optimization

Abdelkader Dib, Milos Balac

Center for Sustainable Future Mobility (CSFM), ETH Zürich, Switzerland

Correspondence: dabdelkader@ethz.ch

Abstract Calibrating agent-based transport models is usually treated as a black-box problem that needs hundreds to thousands of independent simulation runs. We present an alternative, single-run calibration framework in which parameter updates are embedded directly into the MATSim iteration loop, drawing on stochastic batch optimization from machine learning. Choice variables logged during each MATSim iteration are reused as an on-the-fly dataset for a derivative-free optimizer, and updates are smoothed with an exponential moving average. We illustrate the approach on a large-scale Switzerland scenario, including a validation at the level of individual cantons, where mode-specific constants are calibrated separately for each of the 26 cantons within a single run.

Keywords Agent-based modeling; calibration; optimization; transportation modeling; MATSim.

1 Introduction

Agent-based models (ABMs) are widely used in transportation planning because they can represent the heterogeneous behavior of individual travelers and their interactions [1, 2], capturing responses to policy interventions and new mobility solutions that aggregate models cannot [3, 4]. A persistent obstacle to their practical use is calibration [2, 5, 6]: adjusting model parameters so that simulated outputs match observed data.

In MATSim [7], mode choice is governed by utility functions whose parameters, alternative-specific constants (ASCs) and marginal utilities, are typically estimated from survey data and then calibrated, either manually [8–11] or with black-box optimization [12, 13]. Both routes require many complete simulation runs; for large-scale scenarios, where a single run can take hours, this quickly becomes the dominant computational cost of a modeling project.

This work presents an alternative paradigm: *online* calibration, where parameter adjustment happens *within* a single simulation run rather than across many independent ones, using an idea borrowed from stochastic optimization in machine learning.

2 Related Work

ABMs are used across many fields: economics [14, 15], sociology [16], biology and ecology [17–19], epidemiology [5], energy systems [20], land use [21], and emissions [22, 23]. Calibration is a shared bottleneck [24]. Beyond manual tuning and grid search, common black-box strategies include evolutionary and population-based methods such as genetic algorithms [25, 26], particle swarm optimization [27], and CMA-ES [12]; Bayesian optimization [28, 29]; and gradient-free stochastic approximation such as SPSA [13, 30]. Surrogate models (Gaussian processes, random forests, neural networks) are also used to reduce the number of full simulation runs [31–33], though at the cost of extra surrogate-validation effort.

Within transportation, Liang [13] applied SPSA to MATSim utility parameters, Hörl [12] built a general calibration framework around CMA-ES, SPSA and Natural Evolution Strategies, Agarwal et al. [34] used the search-acceleration

strategy of Flötteröd [35], and Zhang et al. [36] and Agriesti et al. [37] used surrogates to cut down the number of runs. A more targeted, closed-form update rule for ASCs was proposed by Rakow and Nagel [38] and applied iteratively by Rakow et al. [39], alternating a full MATSim run with an ASC update after each run. All of these methods, however, still rely on repeated, independent full simulations. Our approach instead abandons the multi-run paradigm altogether.

3 Background

Our work is based on the eqasim pipeline [8], which combines census data, travel surveys and OpenStreetMap into a synthetic population with individual attributes and daily activity plans, representative of the travel demand. It provides ready-to-use scenarios such as Île-de-France [40], California [41], and Switzerland [42]. Simulations are run in MATSim [7], an open-source, activity-based transport simulator that iteratively re-plans a subset of agents each iteration and drives the system toward a stochastic user equilibrium. Mode choice is handled by MATSim’s discrete mode choice (DMC) extension [43, 44], in which a subset of agents re-evaluate their mode using a multinomial logit model [45, 46] at every iteration.

Our calibration framework borrows from stochastic gradient (SG) methods in machine learning [47, 48], which update model weights on small, randomly sampled mini-batches rather than on the full dataset at once, with variants such as momentum [49] and Adam [50] smoothing the resulting noisy updates.

4 Methodology: Online, Single-Run Calibration

The framework rests on a *quasi-static assumption*: because only a small share of agents (typically 5%) replan their mode choice at each MATSim iteration, network conditions, and therefore the choice variables (travel time, cost, waiting time, transfers, ...) faced by replanning agents change only marginally between two consecutive iterations, especially after an initial warm-up. This makes it possible to *log* rather than *re-*

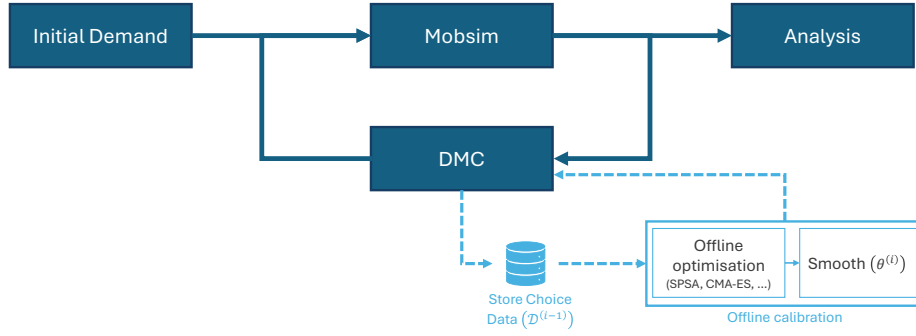


Figure 1: Discrete Mode Choice Calibration Loop: choice variables logged during each MATSim iteration feed a derivative-free optimizer, whose output parameters are fed back into the next iteration.

simulate the decision context: at every iteration i , the choice variables and chosen mode of every replanning agent are recorded, forming a dataset $\mathcal{D}^{(i)}$.

Given $\mathcal{D}^{(i)}$, the deterministic utility and choice probability of any candidate parameter vector θ (ASCs and marginal utilities) can be recomputed offline, instantly, without calling MATSim again. This turns each MATSim iteration into a mini-batch, illustrated in Figure 1: a derivative-free optimizer (we use CMA-ES, though SPSA, PSO, TPE and genetic algorithms are also supported) searches for the θ that minimizes the discrepancy between simulated and observed mode shares (e.g., a log-MSE or Jensen–Shannon divergence loss) and the resulting parameters are fed back into MATSim for the next iteration. Calibration therefore converges alongside the simulation itself, with no restart and no additional full runs. In our experiments, optimizing seven utility parameters with CMA-ES increased the computation time by only 2–3 minutes per MATSim iteration, corresponding to 8–13% of the runtime of a single iteration.

Because updates are computed on a small, randomly varying subset of agents, raw proposals $\theta^{*(i)}$ can be noisy. We stabilize them with an exponential moving average (EMA), analogous to momentum in ML optimizers:

$$\theta^{(i)} = \gamma \theta^{(i-1)} + (1 - \gamma) \theta^{*(i)}, \quad (1)$$

where $\gamma \in [0, 1]$ is a decay factor, typically started low for fast early adaptation and increased toward 1 as the simulation approaches equilibrium.

5 Validation: Calibration of Canton-Specific ASCs

We validated the approach on the eqasim Switzerland scenario [42], a national, 5%-sample, all-canton simulation with five modes (walk, bike, car, car passenger, PT), extensively used in prior work [51]. A single, nationwide ASC per mode reproduces Switzerland’s aggregate mode shares well, but hides substantial regional heterogeneity across the 26 cantons, which differ in language, geography, taxation, and cultural characteristics. To test the proposed methodology and its capability to scale to finer spatial granularity, we replaced the nationwide constant α_j with a canton-specific constant α_{jc} , updated every MATSim iteration from the mode choices of agents residing in canton c only.

Because only 5% of a 5%-sampled population replans per iteration [52], some cantons yield very few replanning observations per iteration. We therefore only update a canton’s constants once at least 800 replanning agents have accumulated for that canton, pooling observations across MATSim iterations for smaller cantons if needed.

Figure 2 shows the evolution of the calibrated parameters over MATSim iterations. During the 90-iteration run, the parameters for most cantons stabilize by approximately iteration 60, which is the same number of iterations required for an uncalibrated run to reach equilibrium. This indicates that cantonal calibration does not require additional iterations. Figure 3 compares the resulting mode shares by canton with those reported in the Swiss Microcensus on Mobility and Transport [53]. The results show good agreement between the simulated and observed mode shares across all simulated cantons.

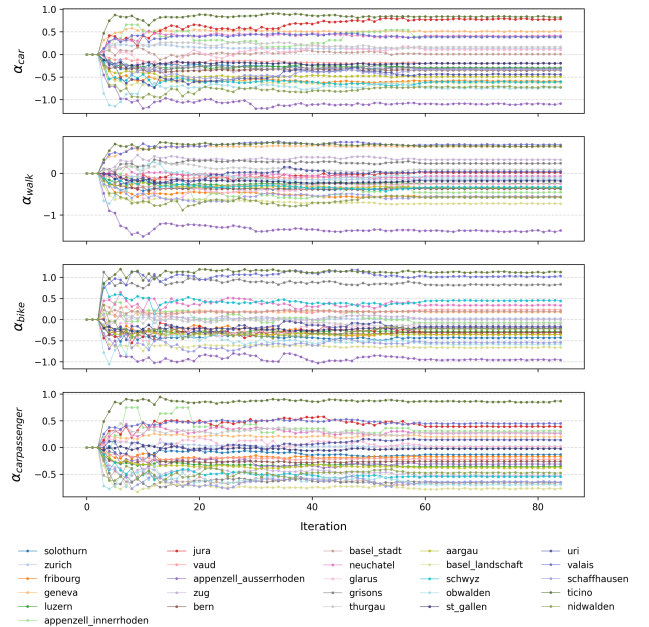


Figure 2: Simulated vs. observed mode shares within each of the 26 Swiss cantons, calibrated online within a single MATSim run.

Achieving this level of spatial granularity with a traditional multi-run pipeline would require hundreds or thousands of independent full simulations, whereas the proposed

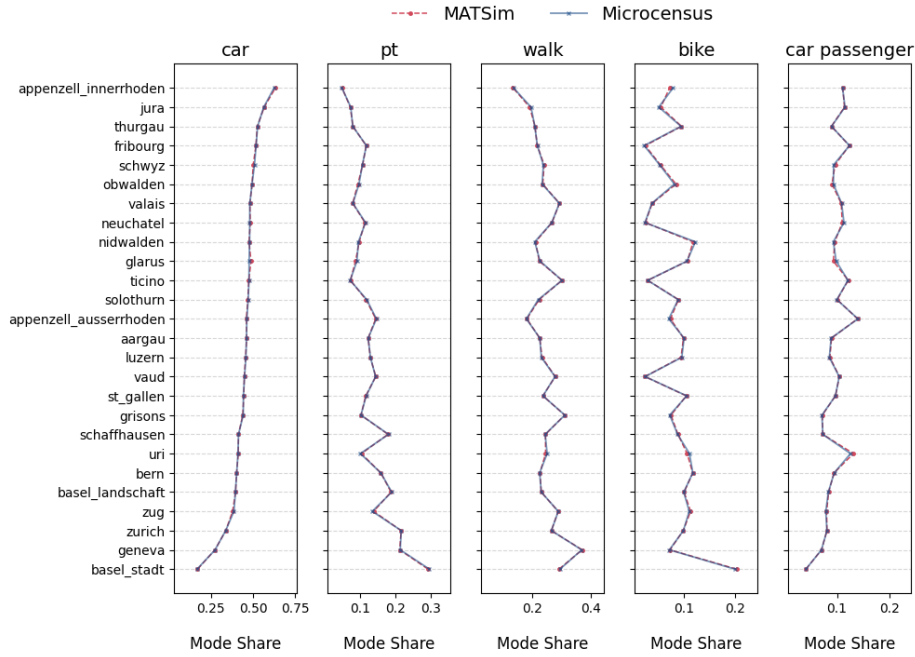


Figure 3: Simulated vs. observed mode shares within each of the 26 Swiss cantons, calibrated online within a single MATSim run.

approach incurs only a small additional computational cost.

6 Discussion and Outlook

Online calibration converges in roughly the same number of iterations as an uncalibrated run, at a marginal computational overhead. As with any non-convex calibration problem [6], multiple local minima remain a concern, and momentum smoothing can occasionally settle parameters in a mixture of local optima rather than a single one; we recommend testing different optimizer and objective function settings (e.g., larger CMA-ES populations, warm-up optimizer from previous runs, multiple objective functions) and, when individual survey observations are available, complementing this framework with discrete choice re-estimation tools such as Biogeme [54]. Natural extensions of this work include applying the same principle to network parameters (capacities, free-flow speeds), to MATSim scoring loop, and to other large-scale ABMs.

Data and Code Availability

The calibration framework is implemented in `eqasim-java`¹; the optimizers are available in Python²; the Switzerland scenario³ and the openly accessible Île-de-France scenario⁴ are both compatible with this methodology.

¹<https://github.com/eqasim-org/eqasim-java/tree/cmdp>

²<https://github.com/Dib-AEK/eqasimParametersCalibration>

³<https://github.com/eqasim-org/eqasim-switzerland>

⁴<https://github.com/eqasim-org/ile-de-france>

References

- [1] Eric Bonabeau. Agent-based modeling: Methods and techniques for simulating human systems. *Proceedings of the national academy of sciences*, 99(suppl_3):7280–7287, 2002.
- [2] Faza Fawzan Bastarianto, Thomas O Hancock, Charisma Farheen Choudhury, and Ed Manley. Agent-based models in urban transportation: review, challenges, and opportunities. *European Transport Research Review*, 15(1):19, 2023.
- [3] Sebastian Hörl, Milos Balac, and Kay W Axhausen. Dynamic demand estimation for an AMoD system in Paris. In *2019 IEEE Intelligent Vehicles Symposium (IV)*, pages 260–266. IEEE, 2019.
- [4] Henrik Becker, Milos Balac, Francesco Ciari, and Kay W Axhausen. Assessing the welfare impacts of Shared Mobility and Mobility as a Service (MaaS). *Transportation research part A: policy and practice*, 131:228–243, 2020.
- [5] Connor Robertson, Cosmin Safta, Nicholson Collier, Jonathan Ozik, and Jaideep Ray. Advancing calibration for stochastic agent-based models in epidemiology with Stein variational inference and Gaussian process surrogates. *arXiv preprint arXiv:2502.19550*, 2025.
- [6] Paul Windrum, Giorgio Fagiolo, and Alessio Moneta. Empirical validation of agent-based models: Alternatives and prospects. *Journal of Artificial Societies and Social Simulation*, 10(2):8, 2007.
- [7] Andreas Horni, Kai Nagel, and Kay W Axhausen. *The Multi-Agent Transport Simulation MATSim*. Ubiquity Press, 2016.
- [8] Sebastian Hörl and Milos Balac. Introducing the eqasim pipeline: From raw data to agent-based transport simulation. *Procedia Computer Science*, 184:712–719, 2021.
- [9] Dominik Ziemke, Ihab Kaddoura, and Kai Nagel. The MATSim Open Berlin Scenario: A multimodal agent-based transport simulation scenario based on synthetic demand modeling and open data. *Procedia computer science*, 151:870–877, 2019.
- [10] Abdelkader Dib, Milos Balac, and Antonio Sciarretta. Deep learning approach to predict microscopic pollutant emissions from mesoscopic traffic simulations. *Transportation Research Part D: Transport and Environment*, 146:104791, 2025.

- [11] Sapan Tiwari, Chengqi Lu, and Afshin Jafari. Network free flow speed calibration in matsim with google api data: A melbourne study. In *Australasian Transport Research Forum*, 2024.
- [12] Sebastian Hörl. Exploring accelerated evolutionary parameter search for iterative large-scale transport simulations in a new calibration testbed. In *10th symposium of the European Association for Research in Transportation (hEART 2022)*, 2022.
- [13] Hong Liang. Calibration of Agent-based Transport Simulation with SPSA method. 2018.
- [14] Shu-Heng Chen, Chia-Ling Chang, and Ye-Rong Du. Agent-based economic models and econometrics. *The Knowledge Engineering Review*, 27(2):187–219, 2012.
- [15] Robert L Axtell and J Doynne Farmer. Agent-based modeling in economics and finance: Past, present, and future. *Journal of Economic Literature*, 63(1):197–287, 2025.
- [16] Michael W Macy and Robert Willer. From factors to actors: Computational sociology and agent-based modeling. *Annual review of sociology*, 28(1):143–166, 2002.
- [17] Gary An, Qi Mi, Joyeeta Dutta-Moscato, and Yoram Vodovotz. Agent-based models in translational systems biology. *Wiley Interdisciplinary Reviews: Systems Biology and Medicine*, 1(2):159–171, 2009.
- [18] Mohammad Soheilypour and Mohammad RK Mofrad. Agent-based modeling in molecular systems biology. *BioEssays*, 40(7):1800020, 2018.
- [19] Adam J McLane, Christina Semeniuk, Gregory J McDermid, and Danielle J Marceau. The role of agent-based models in wildlife ecology and management. *Ecological modelling*, 222(8):1544–1556, 2011.
- [20] Tiejun Ma and Yoshiteru Nakamori. Modeling technological change in energy systems—From optimization to agent-based modeling. *Energy*, 34(7):873–879, 2009.
- [21] Robin B Matthews, Nigel G Gilbert, Alan Roach, J Gary Polhill, and Nick M Gotts. Agent-based land-use models: a review of applications. *Landscape Ecology*, 22(10):1447–1459, 2007.
- [22] Ling Tang, Jiaqian Wu, Lean Yu, and Qin Bao. Carbon emissions trading scheme exploration in China: A multi-agent-based model. *Energy Policy*, 81:152–169, 2015.
- [23] Abdelkader Dib, Antonio Sciarretta, and Milos Balac. Deep learning-based mesoscopic pollutant emissions modeling in road traffic networks. In *2024 Forum for Innovative Sustainable Transportation Systems (FISTS)*, pages 1–7. IEEE, 2024.
- [24] Ana LC Bazzan and Franziska Klügl. A review on agent-based technology for traffic and transportation. *The Knowledge Engineering Review*, 29(3):375–403, 2014.
- [25] Matteo D’Auria, Eric O Scott, Rajdeep Singh Lather, Javier Hilty, and Sean Luke. Assisted parameter and behavior calibration in agent-based models with distributed optimization. In *International Conference on Practical Applications of Agents and Multi-Agent Systems*, pages 93–105. Springer, 2020.
- [26] Enrique Canessa and Sergio Chaigneau. Calibrating agent-based models using a genetic algorithm. *Studies in Informatics and Control*, 24(1):79–90, 2015.
- [27] Alexander Michels, Jeon-Young Kang, and Shaowen Wang. Particle swarm optimization for calibration in spatially explicit agent-based modeling. *Journal of Artificial Societies and Social Simulation*, 25(2), 2022.
- [28] Thomas Lux. Estimation of agent-based models using sequential Monte Carlo methods. *Journal of Economic Dynamics and Control*, 91:391–408, 2018.
- [29] Jakob Grazzini, Matteo G Richiardi, and Mike Tsionas. Bayesian estimation of agent-based models. *Journal of Economic Dynamics and Control*, 77:26–47, 2017.
- [30] Simon Oh, Ravi Seshadri, Carlos Lima Azevedo, and Moshe E Ben-Akiva. Demand calibration of multimodal microscopic traffic simulation using weighted discrete SPSA. *Transportation Research Record*, 2673(5):503–514, 2019.
- [31] Connor Robertson, Cosmin Safta, Nicholson Collier, Jonathan Ozik, and Jaideep Ray. Advancing calibration for stochastic agent-based models in epidemiology with Stein variational inference and Gaussian process surrogates. *arXiv preprint arXiv:2502.19550*, 2025.
- [32] Claudio Angione, Eric Silverman, and Elisabeth Yaneske. Using machine learning as a surrogate model for agent-based simulations. *Plos one*, 17(2):e0263150, 2022.
- [33] Yi Zhang, Zhe Li, and Yongchao Zhang. Validation and calibration of an agent-based model: A surrogate approach. *Discrete Dynamics in Nature and Society*, 2020(1):6946370, 2020.
- [34] Amit Agarwal, Gunnar Flötteröd, and Kai Nagel. Calibration of behavioral parameters in an agent-based transport simulation. 2017.
- [35] Gunnar Flötteröd. A search acceleration method for optimization problems with transport simulation constraints. *Transportation Research Part B: Methodological*, 98:239–260, 2017.
- [36] Chao Zhang, Carolina Osorio, and Gunnar Flötteröd. Efficient calibration techniques for large-scale traffic simulators. *Transportation Research Part B: Methodological*, 97:214–239, 2017.
- [37] Serio Agriesti, Vladimir Kuzmanovski, Jaakko Hollmén, Claudio Roncoli, and Bat-Hen Nahmias-Biran. A Bayesian optimization approach for calibrating large-scale activity-based transport models. *IEEE Open Journal of Intelligent Transportation Systems*, 4:740–754, 2023.
- [38] Christian Rakow and Kai Nagel. Automated calibration of alternative specific con-constants in agent based transport models. 2024.
- [39] Christian Rakow, Moritz Kreuschner, and Kai Nagel. Advancing the MATSim Open Berlin Scenario: Improvements in transport scenario generation and calibration methods. *Transportation Research Procedia*, 86:732–739, 2025.
- [40] Sebastian Hörl and Milos Balac. Synthetic population and travel demand for Paris and Île-de-France based on open and publicly available data. *Transportation Research Part C: Emerging Technologies*, 130:103291, 2021.
- [41] Milos Balac and Sebastian Hörl. Synthetic population for the state of California based on open-data: examples of San Francisco Bay area and San Diego County. In *100th Annual Meeting of the Transportation Research Board (TRB)*, 2021.
- [42] Abdelkader Dib, Aurore Sallard, and Miloš Balac. Agent-based transport model of Switzerland: A fully automated pipeline based on eqasim and MATSim. *26th Swiss Transportation Research Conference, Ascona, Switzerland*, 2026.
- [43] Sebastian Hörl, Miloš Balać, and Kay W Axhausen. Pairing discrete mode choice models and agent-based transport simulation with MATSim. In *2019 TRB Annual Meeting Online*, pages 19–02409. Transportation Research Board, 2019.
- [44] Sebastian Hörl, Milos Balac, and Kay W Axhausen. A first look at bridging discrete choice modeling and agent-based microsimulation in MATSim. *Procedia computer science*, 130:900–907, 2018.
- [45] Kenneth E. Train. *Discrete Choice Methods with Simulation*. Cambridge University Press, Cambridge, UK, 2009.
- [46] Sebastian Hörl. Integrating discrete choice models with MATSim scoring. *Procedia Computer Science*, 184:704–711, 2021.
- [47] Léon Bottou, Frank E Curtis, and Jorge Nocedal. Optimization methods for large-scale machine learning. *SIAM review*, 60(2):223–311, 2018.
- [48] Léon Bottou. Large-scale machine learning with stochastic gradient descent. In *Proceedings of COMPSTAT’2010: 19th International Conference on Computational Statistics Paris France, August 22-27, 2010 Keynote, Invited and Contributed Papers*, pages 177–186. Springer, 2010.

- [49] Goran Nakerst, John Brennan, and Masudul Haque. Gradient descent with momentum—to accelerate or to super-accelerate? *arXiv preprint arXiv:2001.06472*, 2020.
- [50] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [51] Sebastian Hörl, Felix Becker, and Kay W Axhausen. Simulation of price, customer behaviour and system impact for a cost-covering automated taxi system in Zurich. *Transportation Research Part C: Emerging Technologies*, 123:102974, 2021.
- [52] Carlos Llorca and Rolf Moeckel. Effects of scaling down the population for agent-based traffic simulations. *Procedia Computer Science*, 151:782–787, 2019.
- [53] BFS, 2015. Mikrozensus Mobilität und Verkehr (MZMV). <https://www.bfs.admin.ch/bfs/de/home/statistiken/mobilitaet-verkehr/erhebungen/mzmv.html>, 2025. Accessed: 25 August 2025.
- [54] Michel Bierlaire. A short introduction to Biogeme. *Transport and Mobility Laboratory, ENAC, EPFL*, 2023.